SMITH ENGINEERING
Queen's University

Mechatronics and Robotics Engineering

Calculation Sheet

Project: SAMI
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Area: Laptop Buck Converter
Subject: Electrical Design

No.
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Checked by: Hannah Vibien
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Sheet 1 of 1

V_{out} = Laptop voltage = 20V
I_{laptop} = 2.25 A
V_{in} = husky supply = 24V

$$I_{in,husky} = \frac{V_{out} I_{laptop}}{V_{in}\, \eta}$$
$$= \frac{20(2.25)}{24(0.9)}$$
$$= 2.08\ A$$

Assume 90% efficiency buck converter

2.08A < 5A

∴ Safe to use buck converter

Minimum converter current
$I_{converter} = I_{laptop} \times 1.25$
$= 2.25 \times 1.25$
$= 2.8125A$

Add 25% safety buffer (NEC: 125% continuous load rule)

Look for buck converter rated for ≥ 3A